diagnose_cross_validation_partitions.R

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`diagnose_cross_validation_partitions`
Make Cross-Validation Partition Diagnostics

Description

Calculates full-model and fold-training counts from location assignments and aligned binary community responses for feasibility assessment.

Usage

```
diagnose_cross_validation_partitions(
  data_locations = NULL,
  data_assignments = NULL,
  data_community_matrix = NULL,
  cv_strategy = NULL,
  min_taxon_locations = 1L,
  min_taxon_samples = 1L,
  include_full_model = TRUE
)
```

Arguments

`data_locations` Location table returned by [build_cross_validation_location_table()].

`data_assignments`
Assignment table returned by a cross-validation assignment helper. May be empty when ‘cv_strategy = "none"‘ and only full-model diagnostics are required.

`data_community_matrix`
Numeric binary matrix with one row per original sample row and one column per taxon. Row positions must match location ‘row_indices‘.

`cv_strategy` Character scalar identifying the candidate strategy. Either ‘"spatially_stratified_group_kfold"‘, ‘"leave_one_location_out"‘, or ‘"none"‘ for full-model-only diagnostics.

`min_taxon_locations`
Positive integer minimum number of training locations with a taxon presence.

`min_taxon_samples`
Positive integer minimum number of training samples with a taxon presence.

`include_full_model`
Logical scalar. If ‘TRUE‘, prepend the full-model diagnostic row. Defaults to ‘TRUE‘.

Details

Taxon viability is learned independently within each training partition. A retained binary taxon must meet both occurrence thresholds and contain at least one absence. MEM location counts equal the number of training locations; additional mode-specific MEM checks can raise the threshold in the feasibility assessor.

Value

Tibble matching [resolve_cross_validation_strategy()] input, with one optional full-model row and one row per repeat and fold.

Examples

```
data_locations <-
  tibble::tibble(
    location_id = c("a", "b"),
    n_samples = c(1L, 1L),
    row_indices = list(1L, 2L)
  )
data_assignments <-
  build_leave_one_location_out_assignments(data_locations)
diagnose_cross_validation_partitions(
  data_locations = data_locations,
  data_assignments = data_assignments,
  data_community_matrix = matrix(c(0, 1), ncol = 1L),
  cv_strategy = "leave_one_location_out"
)
```

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